INTRODUCTION TO DEEP LEARNING

PART V – LATEST APPLICATIONS, CODE AND KNOWLEDGE SOURCES

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Machine Learning Group
Institute of Data Science

Schedule

Date	Time	Activity
28.05.2026 Day 1	09:00 - 10:00	Introduction and basics
	10:00 - 10:30	Hands-on I
	10:30 - 10:45	Coffee break
	10:45 - 11:15	Hands-on II
	11:15 - 12:00	Hands-on III (Group Work)
	12:00 - 13:00	Lunch break
	13:00 - 14:00	Advanced concept and Convolutional Neural Network
	14:00 - 14:45	Hands-on IV
	14:45 - 15:15	Recap Day 1
29.05.2026 Day 2	09:00 - 10:00	Deep Generative Models
	10:00 - 10:45	Hands-on III
	10:45 - 11:15	Latest applications, code and knowledge sources
	11:15 - 11:30	Closing

We're here!

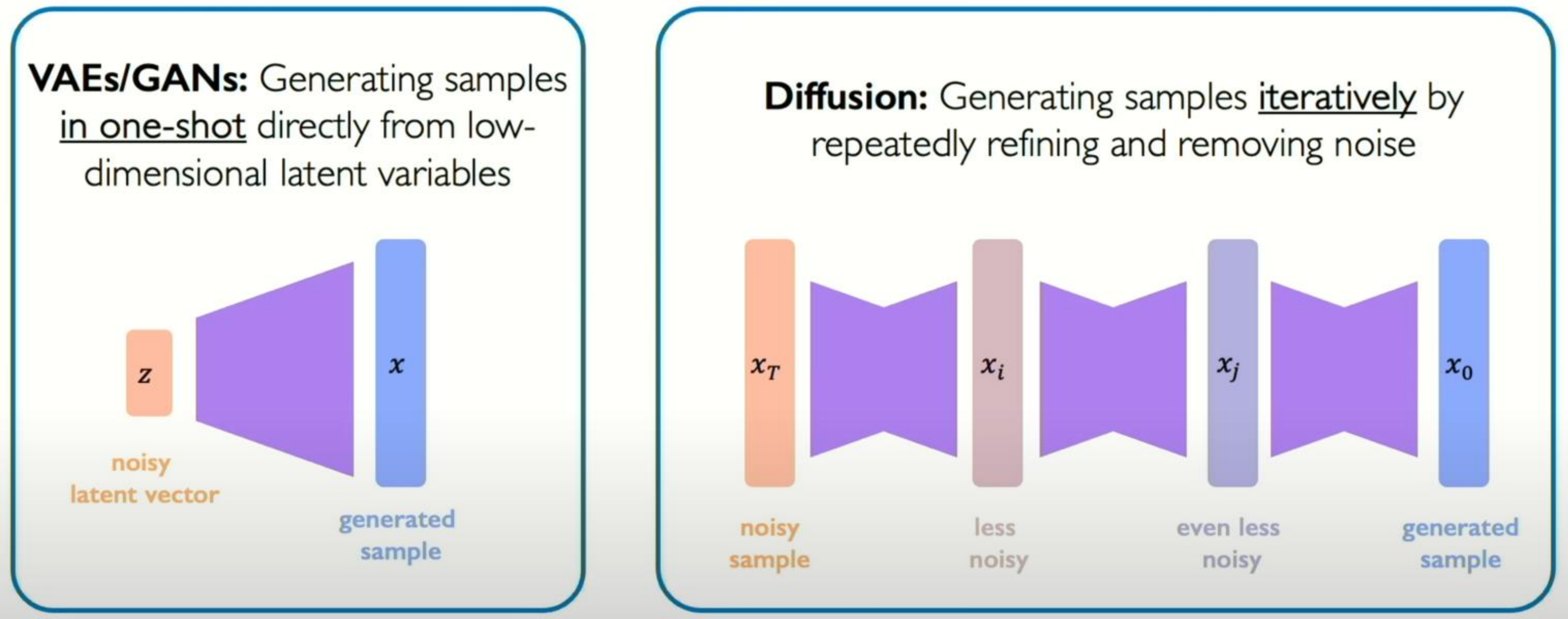
New frontiers

- Transformer
- Diffusion models

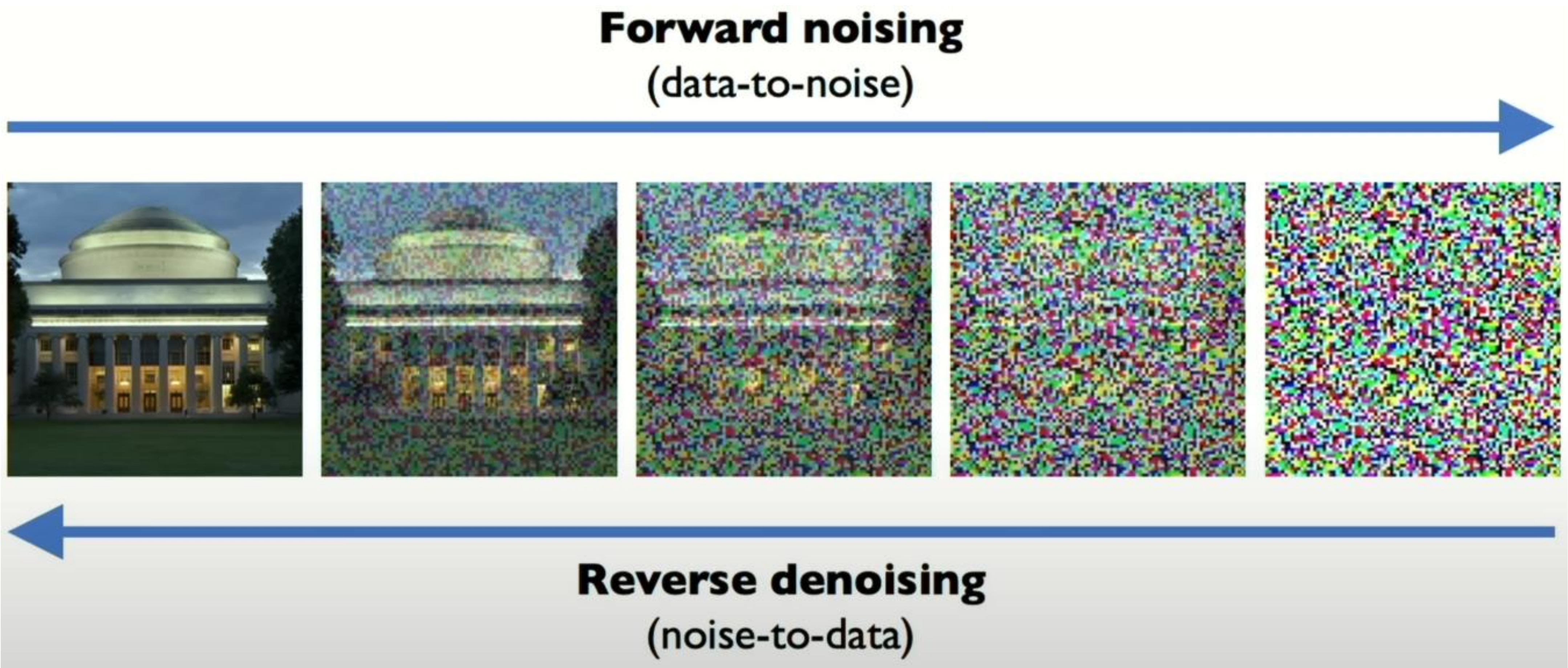
Diffusion model

Generative modeling
Diffusion models

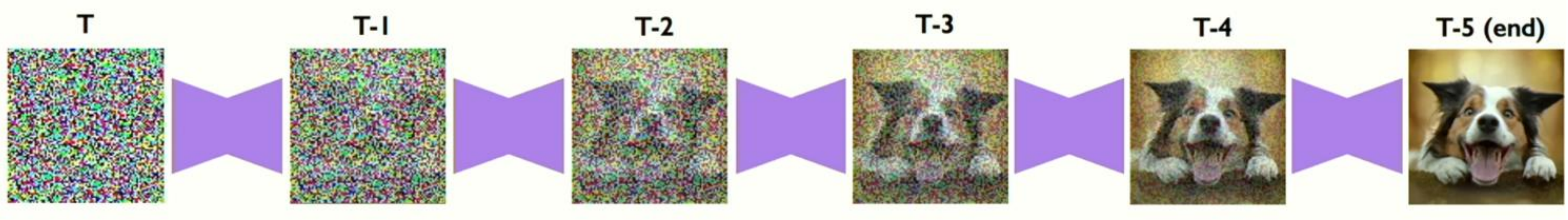
Generative modeling
Diffusion models



Diffusion models
Diffusion process



Diffusion models
Generating new samples



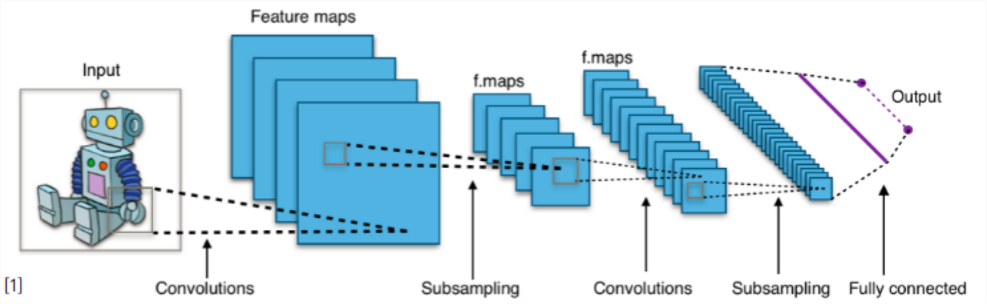
The classic landscape:

One architecture per "community"

Slides from Dr. Lucas Beyer: Intro to Transformers

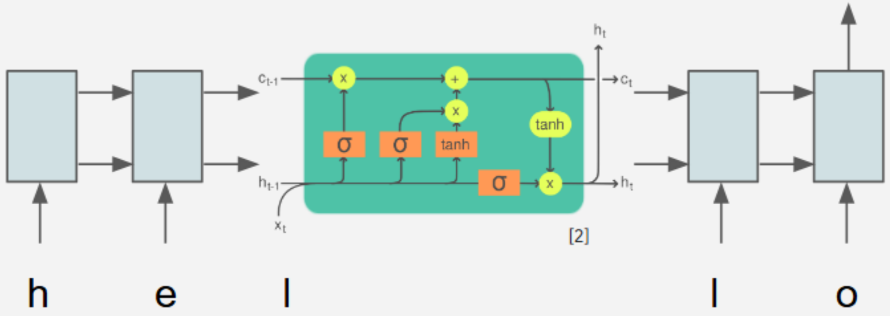
Computer Vision

Convolutional NNs (+ResNets)



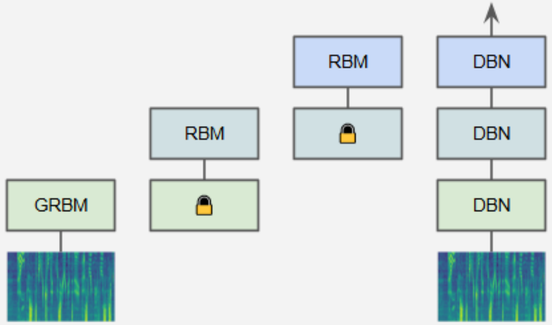
Natural Lang. Proc.

Recurrent NNs (+LSTMs)



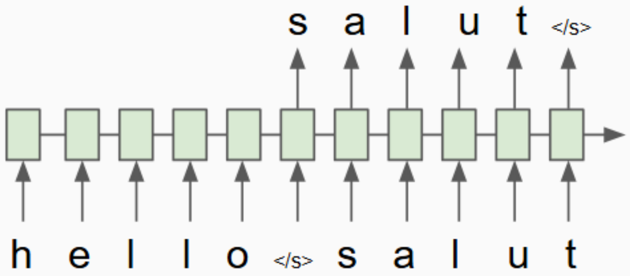
Speech

Deep Belief Nets (+non-DL)



Translation

Seq2Seq



RL

DQN/BC/GAIL

Algorithm 1 Generative adversarial imitation learning

1: **Input:** Expert trajectories $\tau_E \sim \pi_E$, initial policy and discriminator parameters θ_0, w_0
2: **for** $i = 0, 1, 2, \dots$ **do**
3: Sample trajectories $\tau_i \sim \pi_{\theta_i}$
4: Update the discriminator parameters from w_i to w_{i+1} with the gradient

$$\hat{\mathbb{E}}_{\tau_i}[\nabla_w \log(D_w(s,a))] + \hat{\mathbb{E}}_{\tau_E}[\nabla_w \log(1 - D_w(s,a))] \quad (17)$$

5: Take a policy step from θ_i to θ_{i+1}, using the TRPO rule with cost function $\log(D_{w_{i+1}}(s,a))$. Specifically, take a KL-constrained natural gradient step with

$$\hat{\mathbb{E}}_{\tau_i}[\nabla_\theta \log \pi_\theta(a|s) Q(s,a)] - \lambda \nabla_\theta H(\pi_\theta),$$
$$\text{where } Q(\bar{s},\bar{a}) = \hat{\mathbb{E}}_{\tau_i}[\log(D_{w_{i+1}}(s,a)) \mid s_0 = \bar{s}, a_0 = \bar{a}] \quad (18)$$

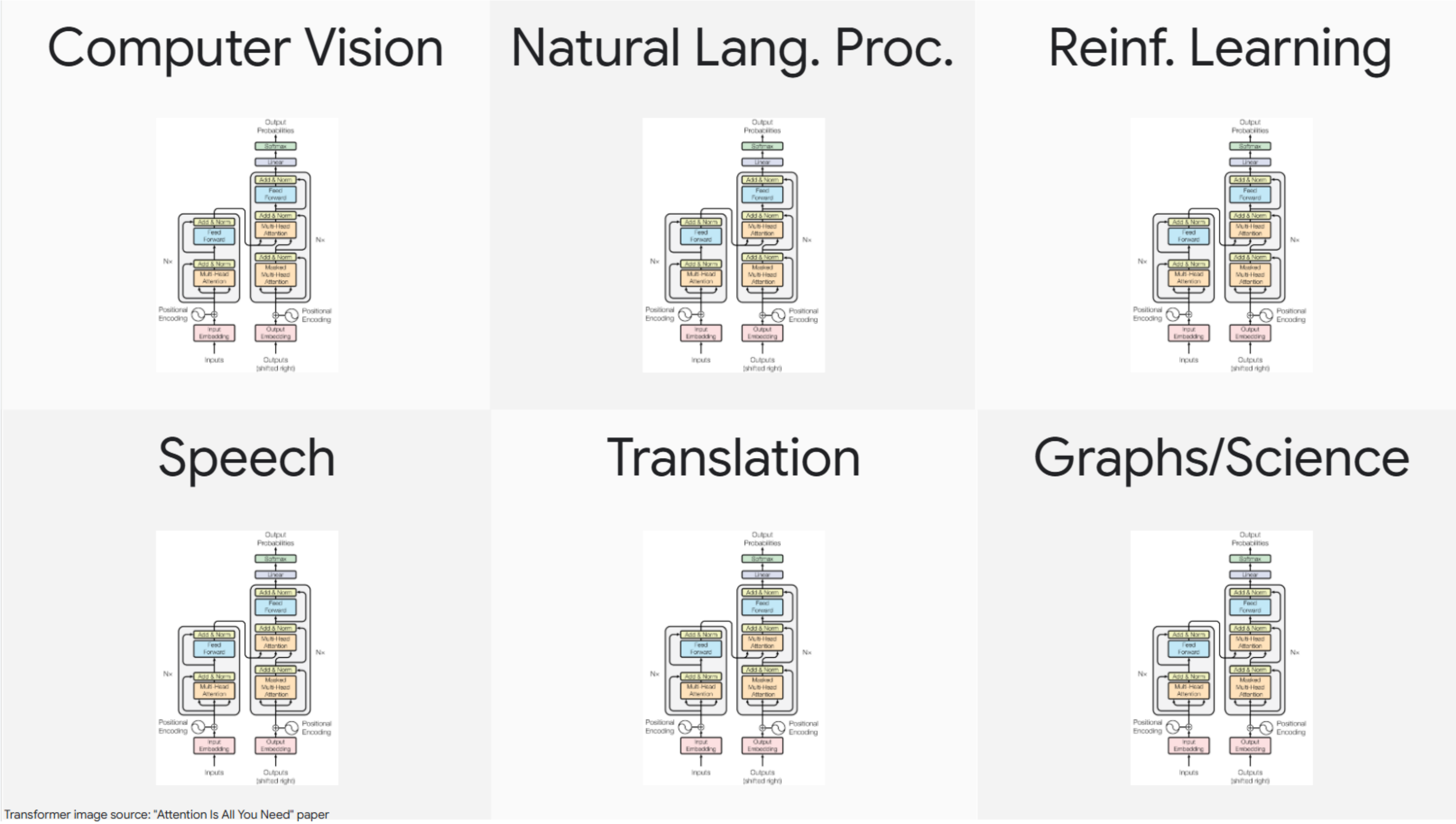
6: **end for**

[1] CNN image CC-BY-SA by Aphex34 for Wikipedia https://commons.wikimedia.org/wiki/File:Typical_cnn.png
[2] RNN image CC-BY-SA by GChe for Wikipedia https://commons.wikimedia.org/wiki/File:The_LSTM_Cell.svg

The Transformer's takeover:

One community at a time

Slides from Dr. Lucas Beyer: [Intro to Transformers]

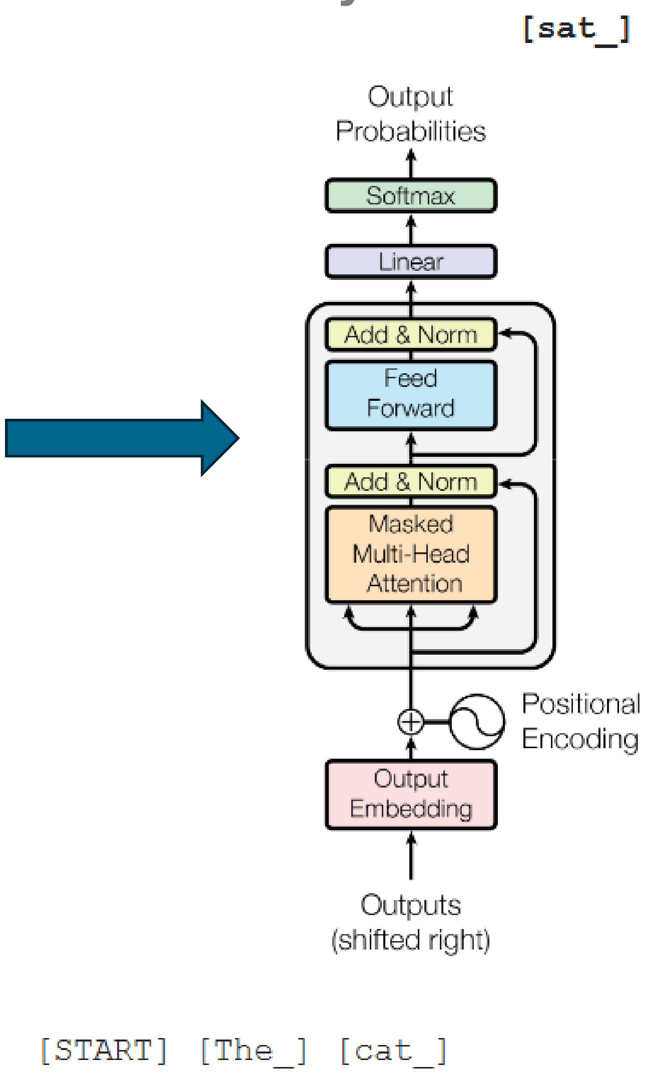


Slides from Dr. Lucas Beyer: [Intro to Transformers]

Slides from Dr. Lucas Beyer: [Intro to Transformers]

Decoder-only GPT

Transformer image source: "Attention Is All You Need" paper



Slides from Dr. Lucas Beyer: Intro to Transformers

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

2020, A Dosovitskiy, L Beyer, A Kolesnikov, D Weissenborn, X Zhai, T Unterthiner, M Dehghani, M Minderer, G Heigold, S Gelly, J Uszkoreit, N Houlsby

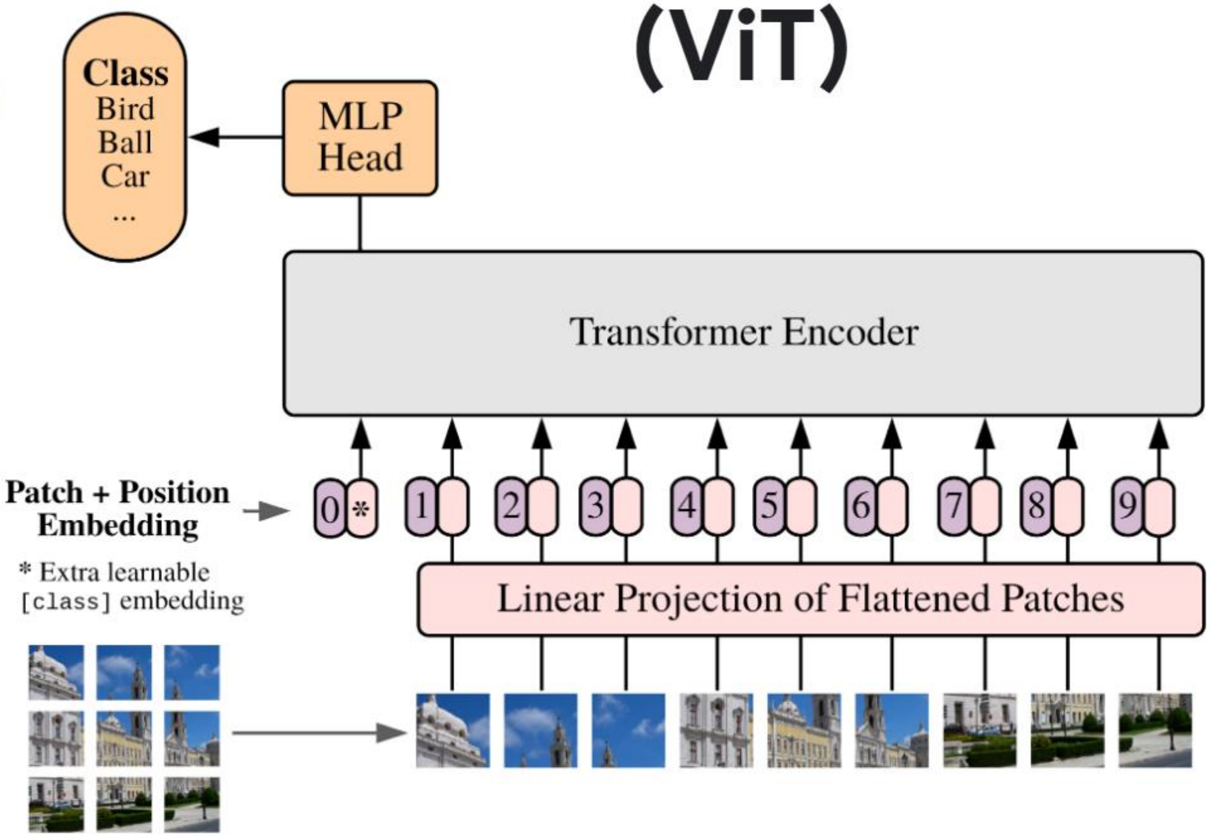
Vision Transformer (ViT)

Many prior works attempted to introduce self-attention at the pixel level.

For 224px², that's 50k sequence length, too much!

Thus, most works restrict attention to local pixel neighborhoods, or as high-level mechanism on top of detections.

The **key breakthrough** in using the full Transformer architecture, standalone, was to **"tokenize" the image** by **cutting it into patches** of 16px², and treating each patch as a token, e.g. embedding it into input space.

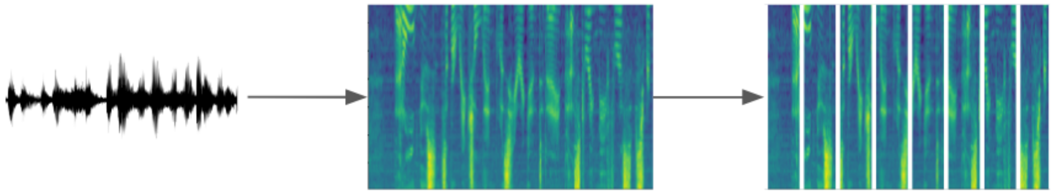


Slides from Dr. Lucas Beyer: Intro to Transformers

Whisper: Robust Speech Recognition via Large-Scale Weak Supervision

2022, A Radford, J W Kim, T Xu, G Brockman, C McLeavey, I Sutskever

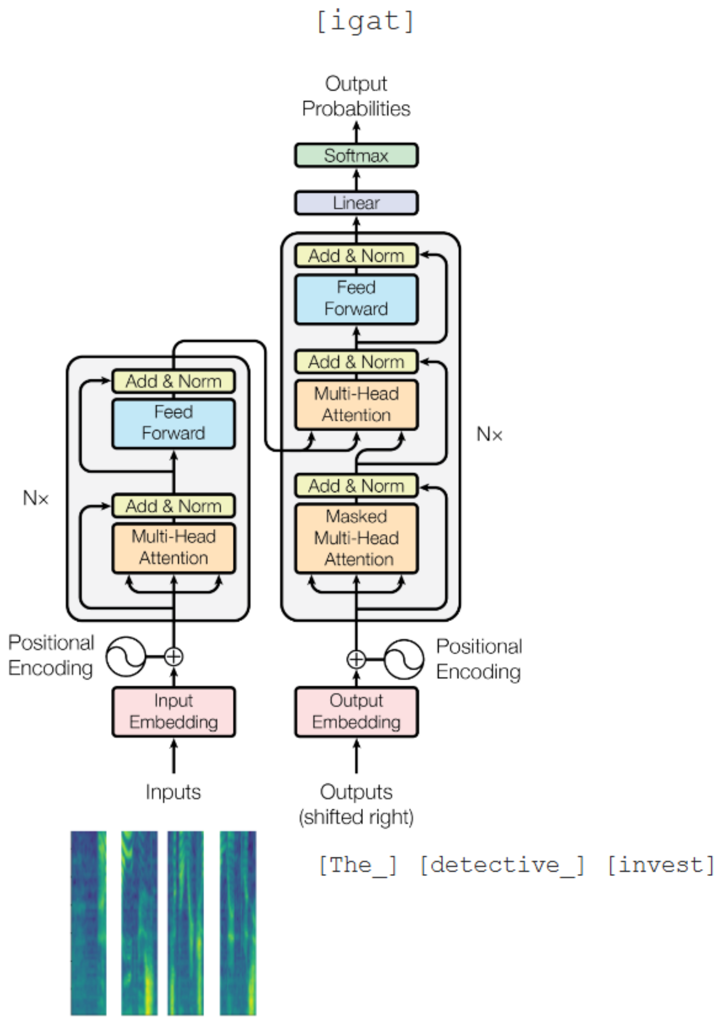
Largely the same story as in computer vision.
But with spectrograms instead of images.



Conformer adds a third type of block using convolutions, and slightly reorder blocks, but overall very transformer-like.

Exists as encoder-decoder variant, or as encoder-only variant with CTC loss.

UPDATE Sept 2022: Plain transformer model in “Whisper” by
A Radford, J W Kim, T Xu, G Brockman, C McLeavey, I Sutskever

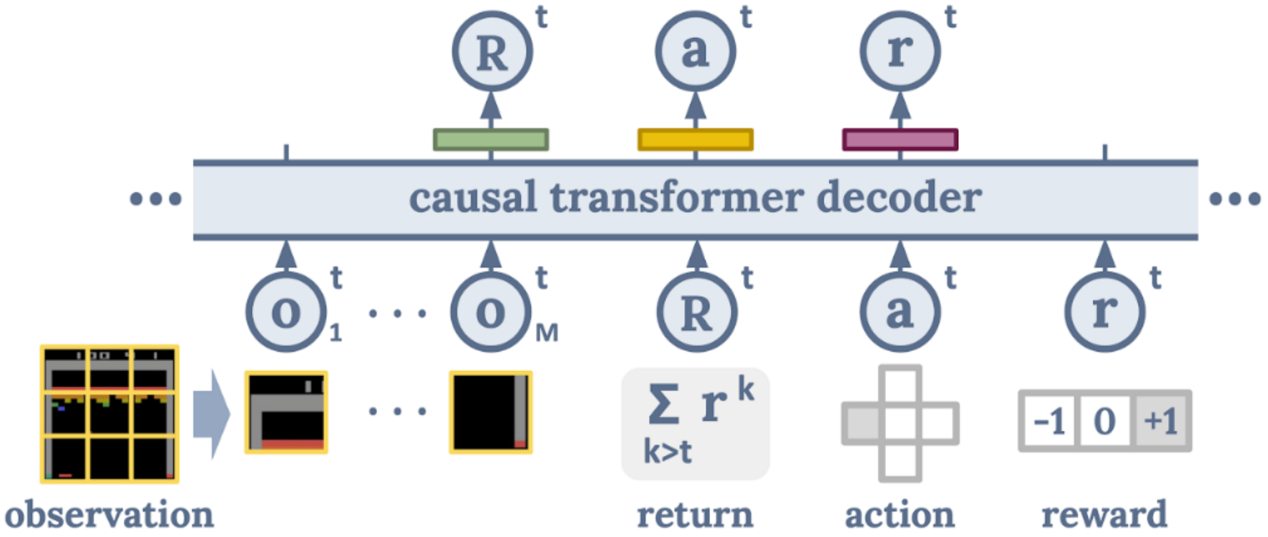


Transformer image source: "Attention Is All You Need" paper

Decision Transformer: Reinforcement Learning via Sequence Modeling

2021, L Chen, K Lu, A Rajeswaran, K Lee, A Grover, M Laskin, P Abbeel, A Srinivas, I Mordatch

Cast the (supervised/offline) RL problem into a sequence ("language") modeling task:



Can generate/decode sequences of actions with desired return (eg skill)

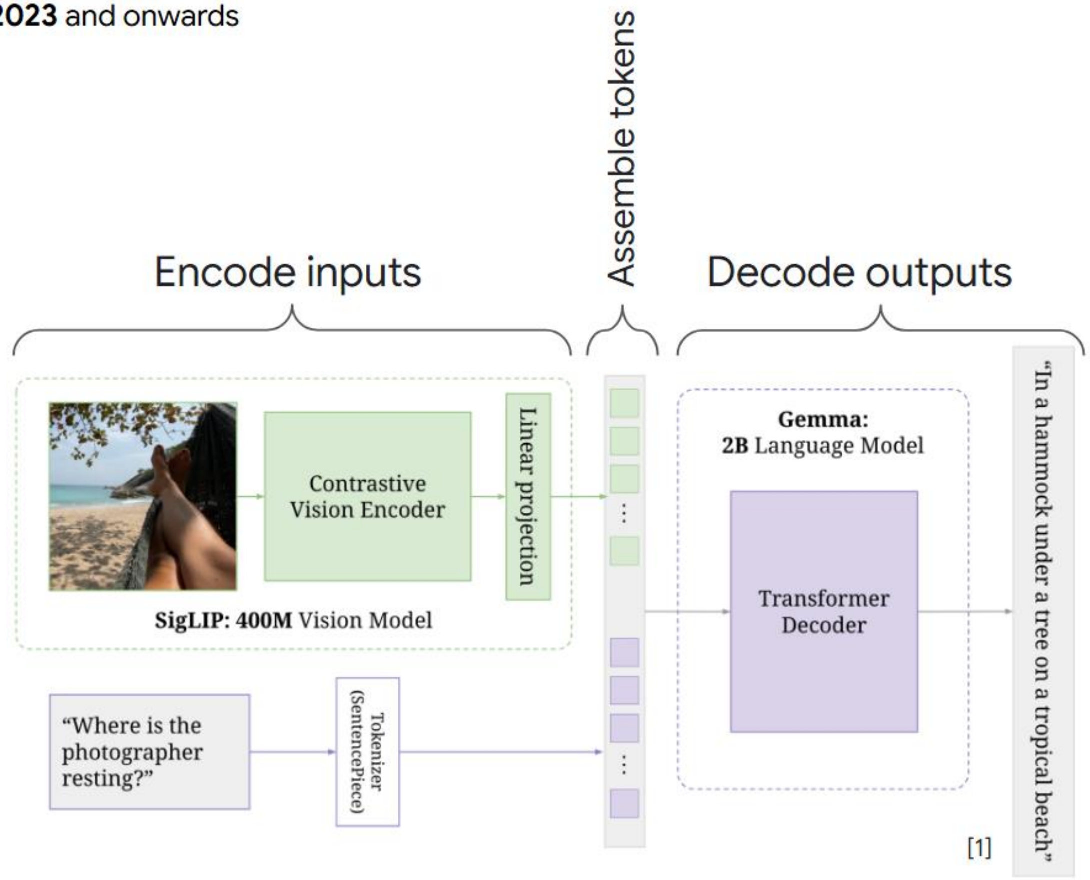
The trick is prompting: `"The following is a trajectory of an expert player: [obs] ..."`

Slide credit: Igor Mordatch

Slides from Dr. Lucas Beyer: Intro to Transformers

Anything you can Generate!

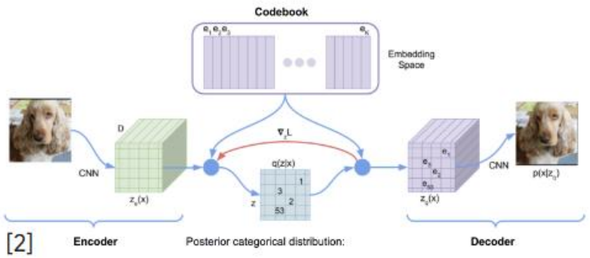
ca 2023 and onwards



[1]

Figure 1 | PaliGemma's architecture: a SigLIP image encoder feeds into a Gemma decoder LM.

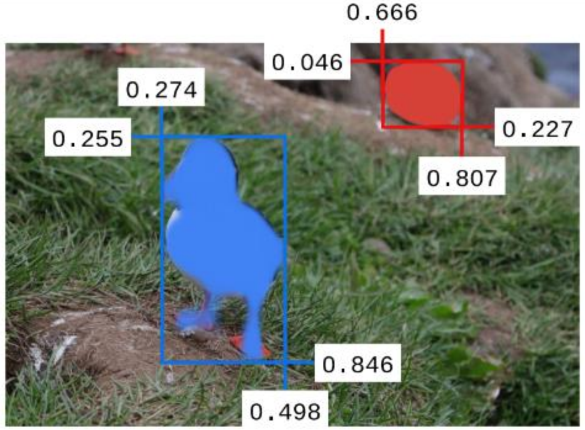
For outputs: learn discrete code (VQ-VAE, FSQ) that represents the output, append to vocab.



[2]

Input:

```
segment puffin in the back ; puffin in front
```



[3]

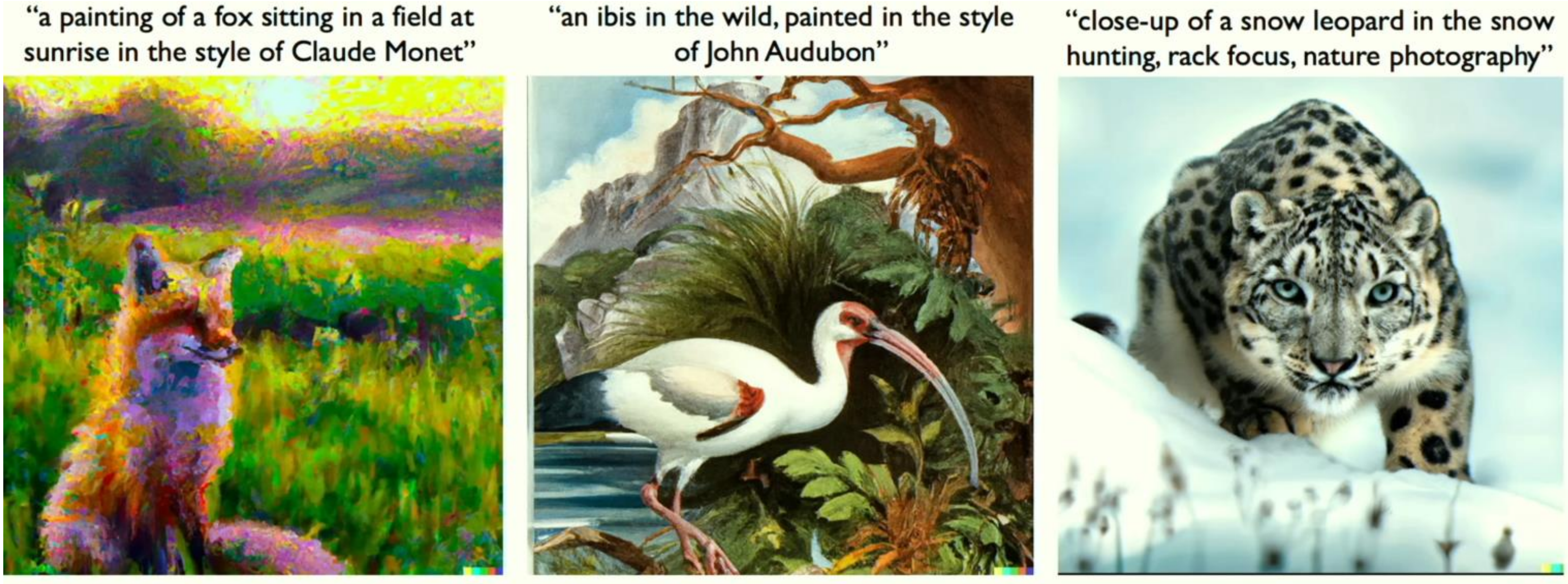
Output:

```
<loc0255><loc0274><loc0846><loc0498>
<seg024>[...]<seg018> puffin in front ;
<loc0046><loc0666><loc0227><loc0807>
<seg106>[...]<seg055> puffin in the back
```

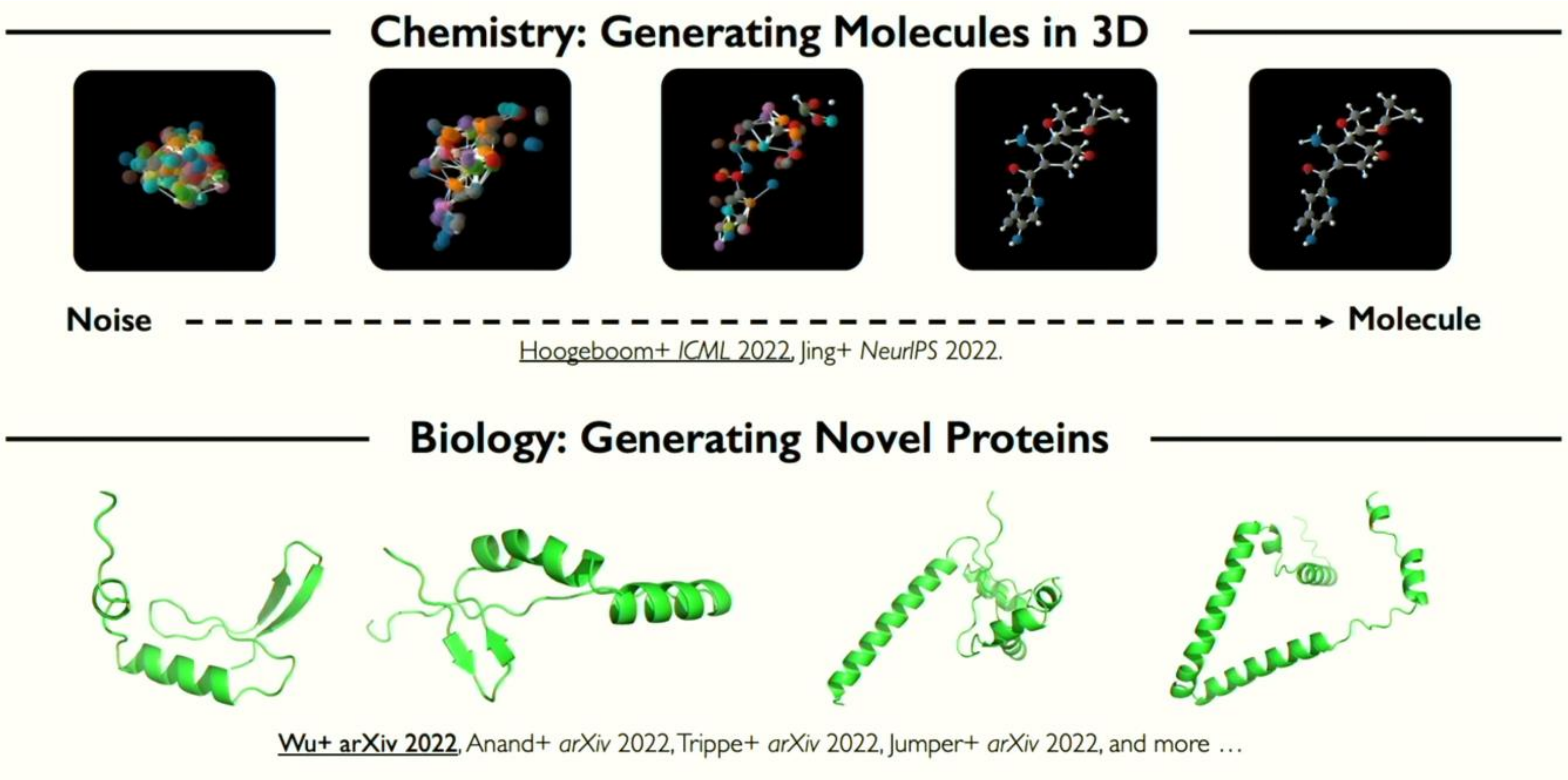

[1] PaliGemma by L. Beyer, A. Steiner, A. S. Pinto, A. Kolesnikov, X. Wang, ..., X. Zhai
[2] VQ-VAE by A. van den Oord, O. Vinyals, K. Kavukcuoglu
[3] PaliGemma 2 by Andreas Steiner, A. S. Pinto, M. Tschannen, ..., L. Beyer, X. Zhai

Slides from Dr. Lucas Beyer: [Intro to Transformers]

Diffusion models
Text to image generation



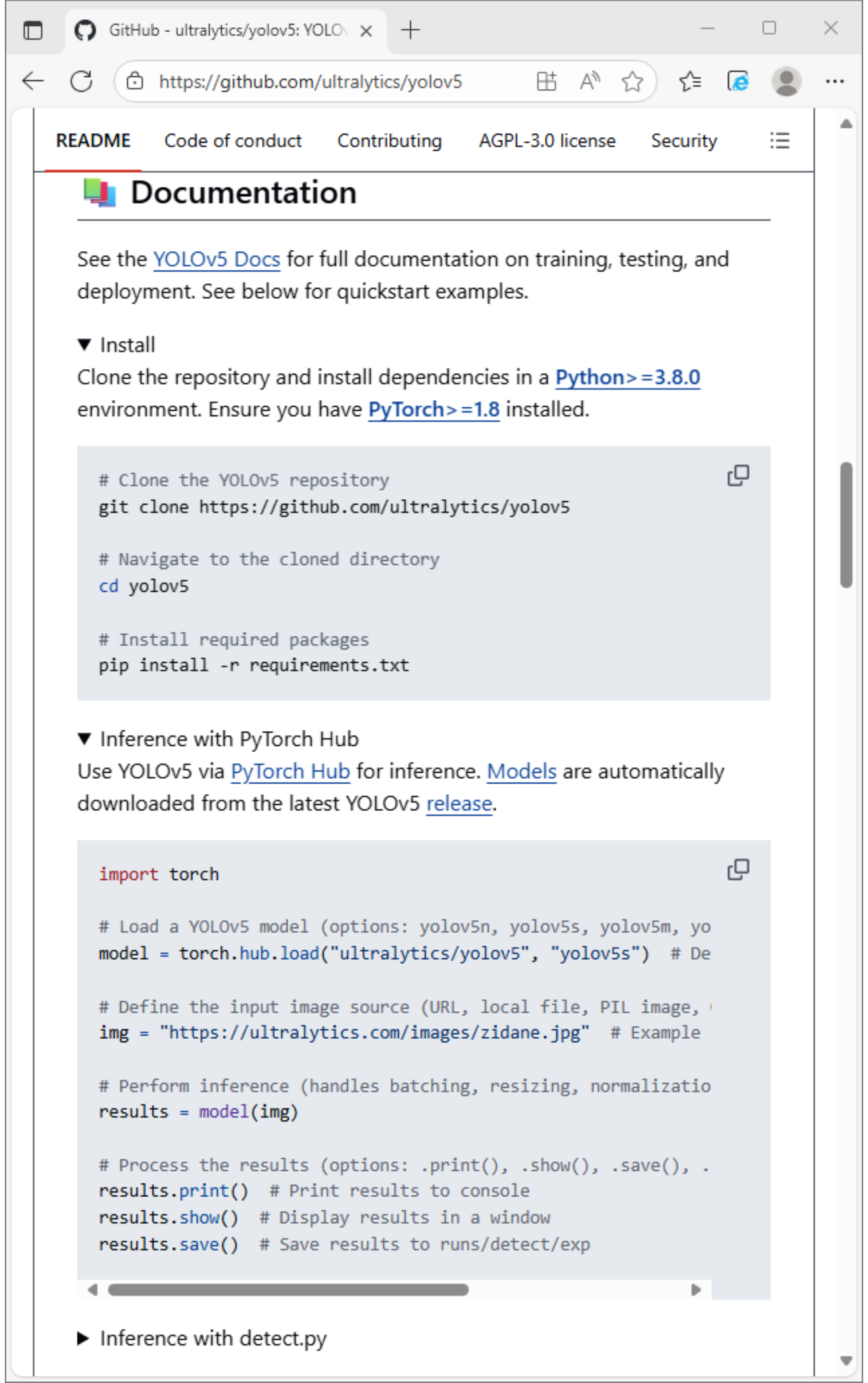
Diffusion models
Beyond images: Molecular Design



Knowledge sources

Code & knowledge sources
Code: Github

- Many free implementations of popular and new methods
- Many conferences now require public code for reproducibility
- Usually a good first step for testing and adapting a new method



GitHub - ultralytics/yolov5: YOLO

https://github.com/ultralytics/yolov5

README | Code of conduct | Contributing | AGPL-3.0 license | Security

Documentation

See the YOLOv5 Docs for full documentation on training, testing, and deployment. See below for quickstart examples.

▼ Install

Clone the repository and install dependencies in a Python>=3.8.0 environment. Ensure you have PyTorch>=1.8 installed.

```
# Clone the YOLOv5 repository
git clone https://github.com/ultralytics/yolov5

# Navigate to the cloned directory
cd yolov5

# Install required packages
pip install -r requirements.txt
```

▼ Inference with PyTorch Hub

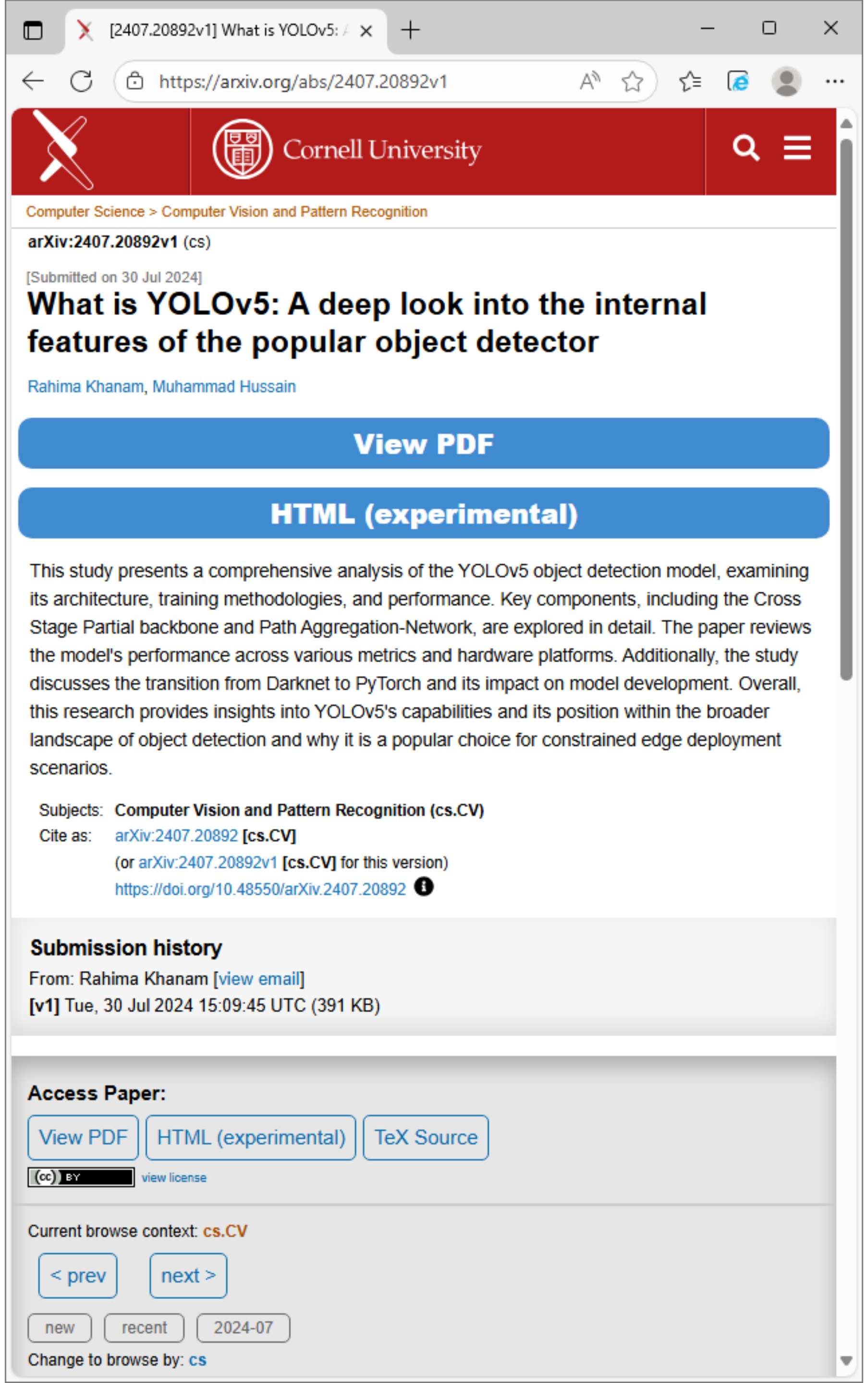
Use YOLOv5 via PyTorch Hub for inference. Models are automatically downloaded from the latest YOLOv5 release.

```
import torch

# Load a YOLOv5 model (options: yolov5n, yolov5s, yolov5m, yo
model = torch.hub.load("ultralytics/yolov5", "yolov5s")  # De

# Define the input image source (URL, local file, PIL image,
img = "https://ultralytics.com/images/zidane.jpg"  # Example

# Perform inference (handles batching, resizing, normalizatio
results = model(img)

# Process the results (options: .print(), .show(), .save(), .
results.print()  # Print results to console
results.show()  # Display results in a window
results.save()  # Save results to runs/detect/exp
```

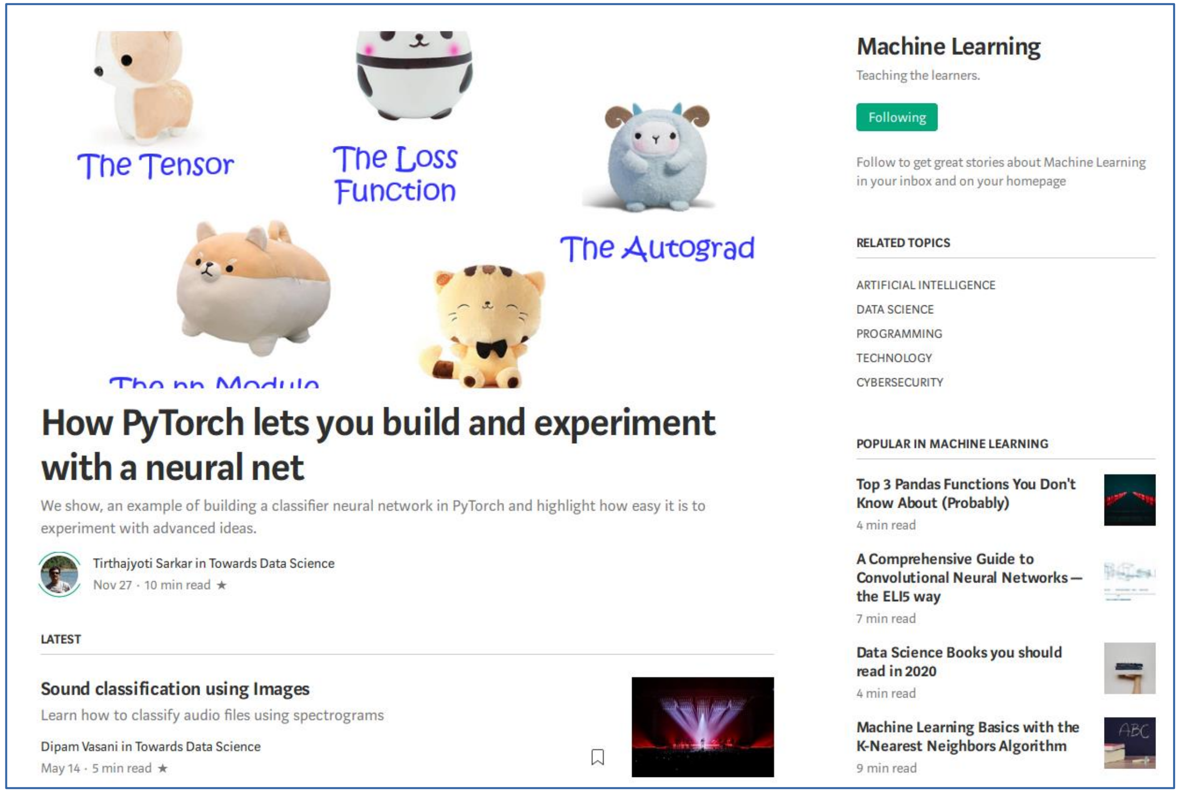
▶ Inference with detect.py

Code & knowledge sources
Knowledge: arXiv

- Now often the first point of publication for DL papers (sometimes the only one)
- Not peer-reviewed!
- Helpful tool:
- http://www.arxiv-sanity.com/



Code & knowledge sources
Knowledge: Blogs


The Tensor
The Loss Function
The Autograd

How PyTorch lets you build and experiment with a neural net

We show, an example of building a classifier neural network in PyTorch and highlight how easy it is to experiment with advanced ideas.

Tirthajyoti Sarkar in Towards Data Science
Nov 27 · 10 min read ★

LATEST

Sound classification using Images
Learn how to classify audio files using spectrograms
Dipam Vasani in Towards Data Science
May 14 · 5 min read ★

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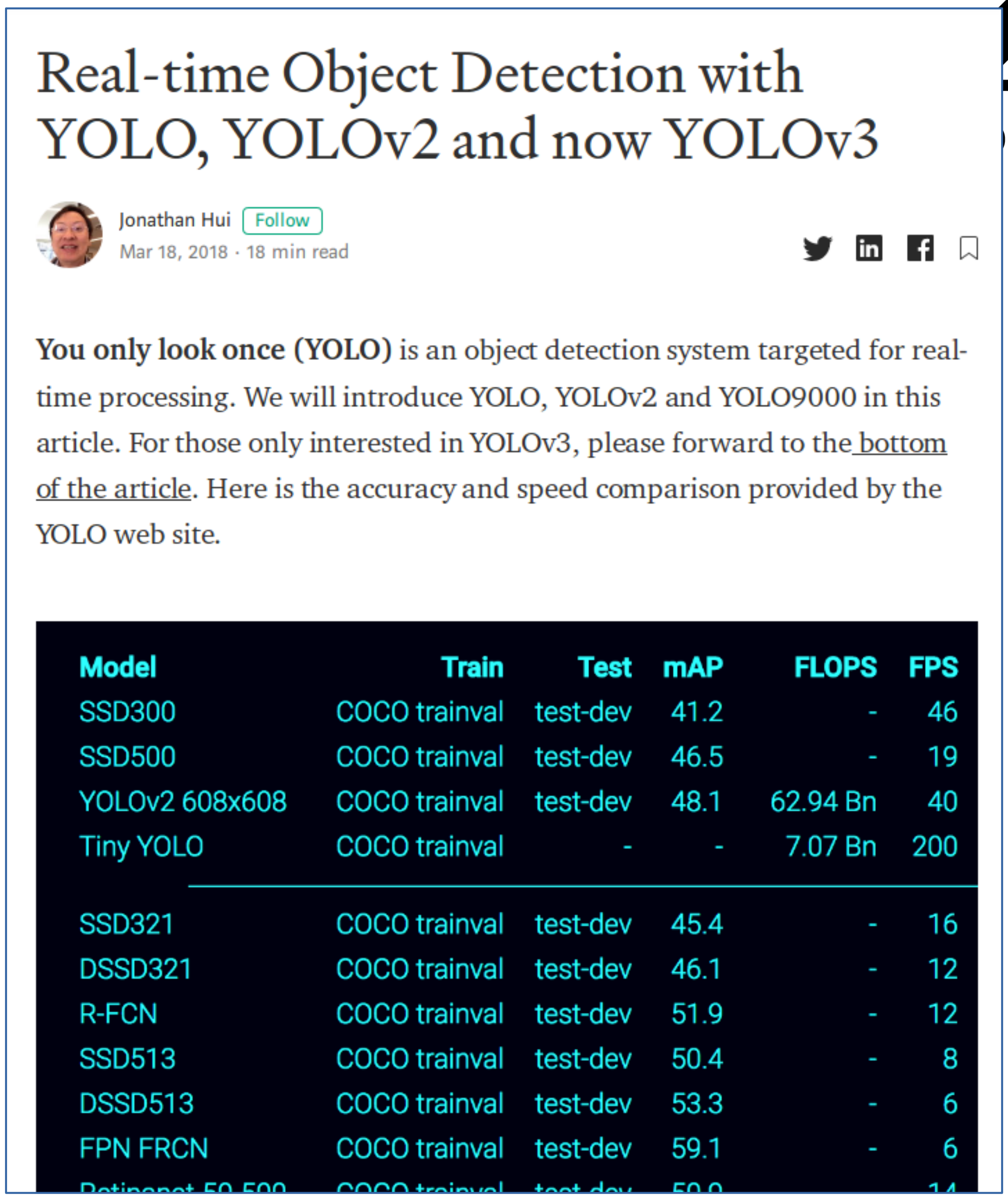
https://medium.com/topic/artificial-intelligence
https://medium.com/topic/machine-learning


Real-time Object Detection with YOLO, YOLOv2 and now YOLOv3

Jonathan Hui Follow
Mar 18, 2018 · 18 min read

You only look once (YOLO) is an object detection system targeted for real-time processing. We will introduce YOLO, YOLOv2 and YOLO9000 in this article. For those only interested in YOLOv3, please forward to the bottom of the article. Here is the accuracy and speed comparison provided by the YOLO web site.

Model	Train	Test	mAP	FLOPS	FPS
SSD300	COCO trainval	test-dev	41.2	-	46
SSD500	COCO trainval	test-dev	46.5	-	19
YOLOv2 608x608	COCO trainval	test-dev	48.1	62.94 Bn	40
Tiny YOLO	COCO trainval	-	-	7.07 Bn	200
SSD321	COCO trainval	test-dev	45.4	-	16
DSSD321	COCO trainval	test-dev	46.1	-	12
R-FCN	COCO trainval	test-dev	51.9	-	12
SSD513	COCO trainval	test-dev	50.4	-	8
DSSD513	COCO trainval	test-dev	53.3	-	6
FPN FRCN	COCO trainval	test-dev	59.1	-	6

Code & knowledge sources
Code: Conferences (selection)

ICML
International Conference
On Machine Learning

NEURAL
INFORMATION
PROCESSING
SYSTEMS

ICLR

IJCAI
International Joint Conferences on
Artificial Intelligence Organization

ROBOTICS
SCIENCE AND SYSTEMS

aaai

CVPR

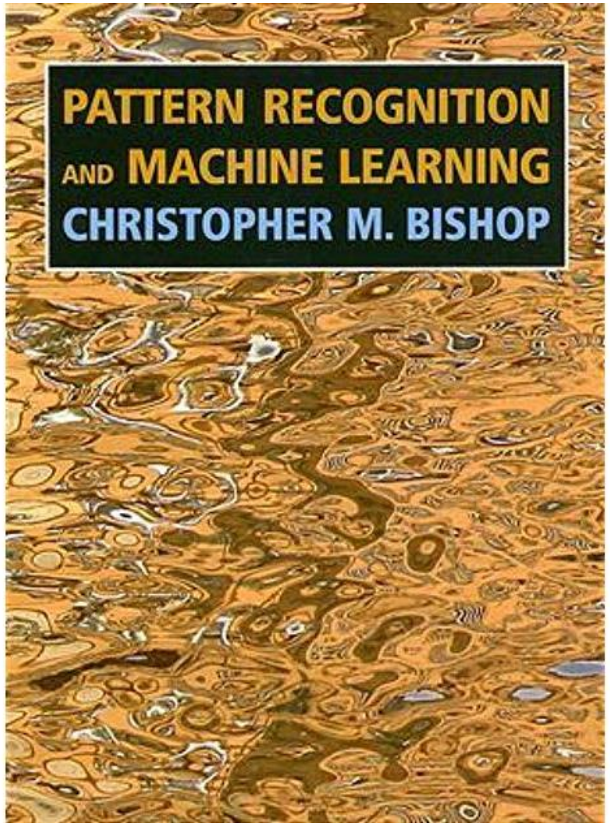
...and of course
many domain-
specific conferences

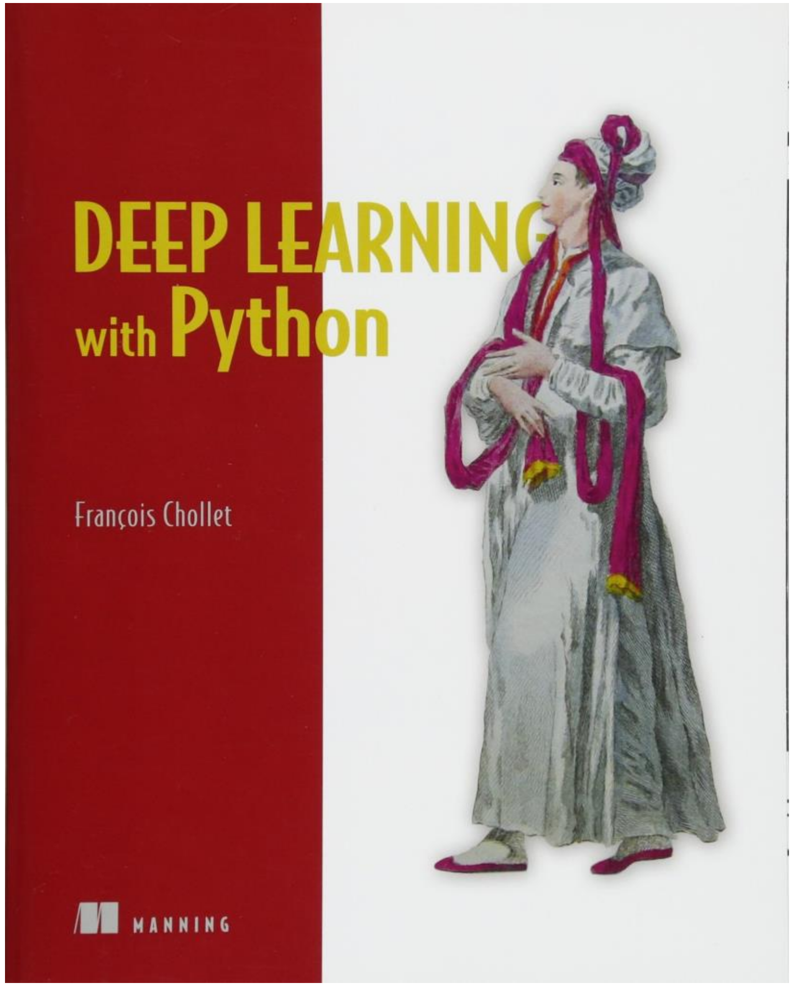
AISTATS

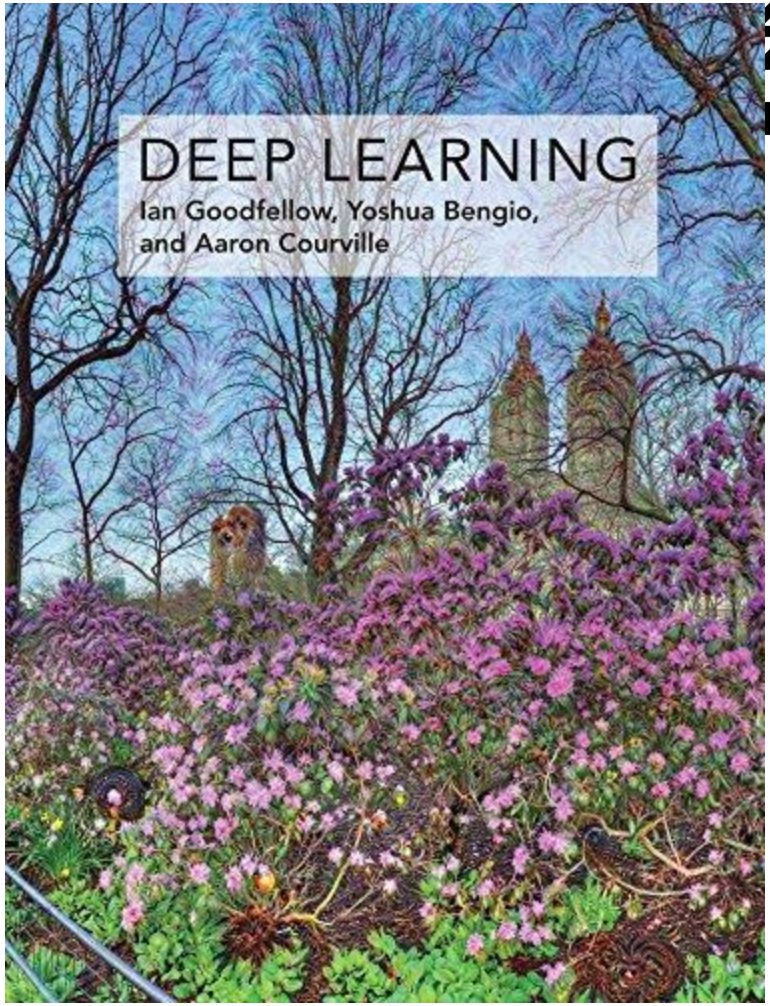
Association for
Computational
Linguistics

EMNLP 2019
Conference on Empirical Methods
in Natural Language Processing

Code & knowledge sources
Knowledge: Books

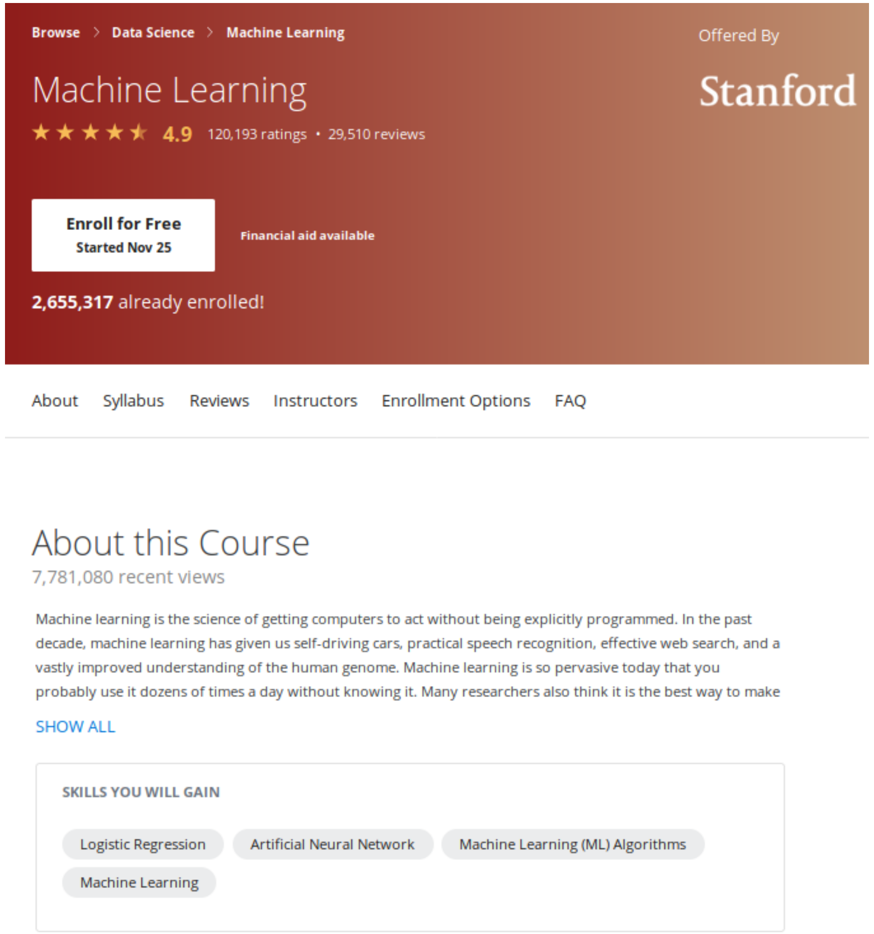


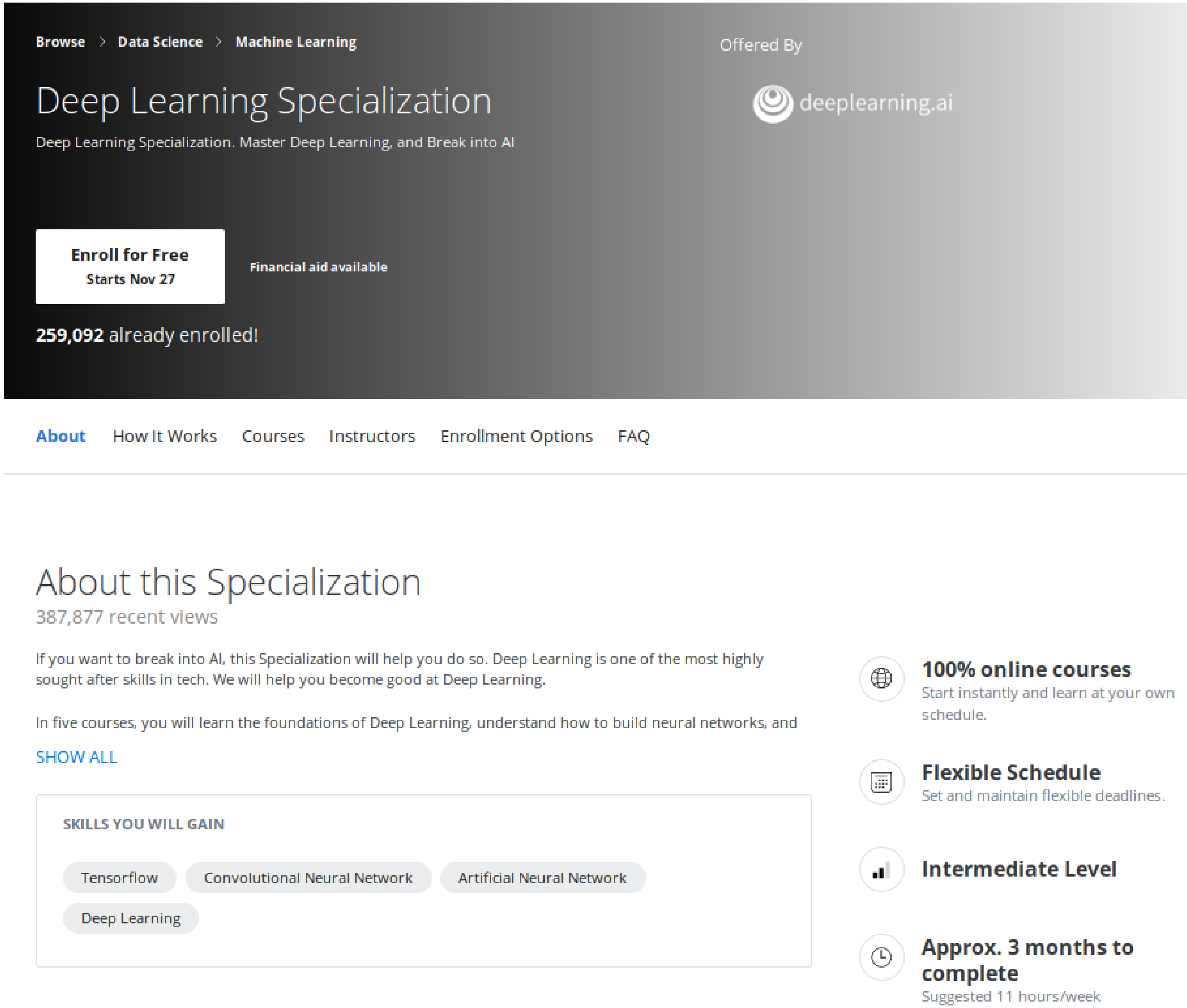




http://www.deeplearningbook.org/

Code & knowledge sources
Knowledge: Online Courses





https://www.coursera.org/

Thank you for your participation!

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Imprint

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Institute:	Data Science
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